

Near-Infrared Absorption and Scattering Separated by **Extended Inverted Signal Correction (EISC)**: Analysis of Near-Infrared Transmittance Spectra of Single Wheat Seeds

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A new extended method for separating, e.g., scattering from absorbance in spectroscopic measurements, extended inverted signal correction (EISC), is presented and compared to **multiplicative signal correction (MSC)** and existing modifications of this. EISC preprocessing is applied to near-infrared transmittance (NIT) spectra of single wheat kernels with the aim of improving the multivariate calibration for protein content by partial least-squares regression (PLSR). The primary justification of the EISC method is to facilitate removal of spectral artifacts and interferences that are uncorrelated to target analyte concentration. In this study, EISC is applied in a general form, including additive terms, multiplicative terms, wavelength dependency of the light scatter coefficient, and simple polynomial terms. It is compared to conventional MSC and derivative methods for spectral preprocessing. Performance of the EISC was found to be comparable to a more complex dual-transformation model obtained by first calculating the second derivative NIT spectra followed by MSC. The calibration model based on EISC preprocessing **performed better than models** based on the raw data, second derivatives, MSC, and MSC followed by second derivatives.

Index Headings: Additive; Multiplicative; Interference; Inverted scatter correction; ISC; Extended inverted signal correction; EISC; Multiplicative signal correction; MSC; Near-infrared transmittance; NIT; Partial least-squares regression; PLSR; Protein; Single seed; Light scattering.

INTRODUCTION

The extended inverted signal correction (EISC) method, originally developed with chemical analyte extensions,¹ is here presented with spectroscopic extensions. This new method is then applied to near-infrared transmittance (NIT) spectra of single wheat kernels prior to multivariate calibration² for protein content, with the calibration model estimated by cross-validated partial least-squares regression (PLSR).^{3,4} The most basic version of EISC, ISC, was originally called “inverted scatter correction” (ISC).⁵ Martens et al.¹ explains the rationale behind the EISC method and its chemically based extensions, in relation to its heritage, the multiplicative signal correction (MSC)^{6,7} and the extended MSC (EMSC).⁸ In the present study, the EISC method is extended with some general physical approximation parameters (wavelength dependency and curvatures), and compared to derivative-based preprocessing.

Today, near-infrared (NIR) spectroscopy in combination with multivariate calibration has become the estab-

lished method for protein determination in cereal breeding as well as for quality determination in the cereal industry, relieving the more than 100-year-old, slow, chemical analysis invented at the Carlsberg Laboratories by the Danish chemist Johan G. Kjeldahl in 1883.⁹ The advantages of using NIR spectroscopic methods for cereal quality are mainly the speed of the analysis and their noninvasive character, which is essential if seed fertility is the aim in breeding programs. As an important spin-off, NIR methods provide possibilities for simultaneous determination of additional quality parameters such as moisture, starch, and fiber content.

In low-cost/high-speed analyses of complex systems such as whole-wheat grain, the pattern of optical paths is very complex, and several physical phenomena may contribute to the apparent pattern of “light scatter”. The information in NIR spectra usually results from both diffuse light scatter and chemically (vibrationally) absorbed light by the sample, and the NIT spectra of single seeds can be considered a worst case with large additive and multiplicative scatter effects due to differences in kernel size, structure, and presentation angle. It is not uncommon to see more than 95% of the variance in NIR $\log(1/T)$ or $\log(1/R)$ data caused by uncontrolled light scattering variations, which usually will dominate the first latent variable in PCA (principal component analysis) or PLSR modeling. In some cases this is desirable, as when the quality to be calibrated for is physical and related to light scattering, e.g., hardness variation of wheat kernels or particle size variation in powders. However, in most cases light scattering creates selectivity and linearity problems for simple quality attributes related to chemical concentrations. In such cases it is imperative that scatter is isolated from the NIT spectra prior to calibration in order to provide a robust and accurate quantitative method.

The single seed protein system has been studied in depth by Delwiche,¹⁰ who found that an optimal data transformation prior to PLSR calibration was obtained by first calculating the second derivative spectra and then correcting them by MSC. The performance of this double transformation model is confirmed by this study,¹¹ but such a complex pre-transformation naturally calls for the development of more general and powerful pre-transformations. In the present study it is demonstrated that a general form of EISC is able to provide a quantitative protein model with a precision equal to Delwiche’s doubly pre-transformed model.

THEORY

In its most basic “ISC” form, the EISC data transformation can correct a combination of additive and multipli-

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cative interference effects in measured spectra, analogous to the original MSC method.^{6,7} Both the MSC and the ISC/EISC adjust the input spectrum of each sample, $\mathbf{z}_i$ in a set of samples, $i = 1, 2, \dots$, towards a common reference spectrum, $\mathbf{m}$, in order to separate possible physical effects from possible chemical absorption effects. The difference between the methods is that the ISC simply reverses regressor and regressand in each sample's regression model between $\mathbf{z}_i$ and $\mathbf{m}$. Like the conventional MSC, the ISC ("basic EISC") preprocessing method estimates and isolates two presumably physical effects for each sample: an *additive* baseline offset effect and a *multiplicative* scaling effect. If the input information $\mathbf{z}_i$ represents absorbance ($A = \log(I_0/I) = \log(1/T)$) values, the additive baseline offset is intended to model an unknown, fixed amount of absorbance lost at every wavelength, e.g., due to light failing to reach the detector because of dispersion of light in the sample. Multiplicative scaling is intended to model an unknown amplification of the absorbance at every wavelength, e.g., due to a change in the effective optical path length because of light scattering effects in the sample.

In the original EISC paper,¹ the basic version of EISC was extended with *chemical* information known *a priori* to represent absorbance spectra from interfering constituents. In the present paper the EISC is instead extended with *physical* information representing wavelength and polynomial extensions when compared to MSC.

Multiplicative Signal Correction. The multiplicative signal correction, originally named **multiplicative scatter correction**, MSC, involves correcting each input spectrum $\mathbf{z}_i = [z_{i1}, z_{i2}, \dots, z_{ik}, \dots, z_{iK}]$ in a set of related samples $i = 1, 2, \dots$ towards an ideal spectrum $\mathbf{m}$ where the influence of physical scattering variations has been removed from the effects of chemical absorbance (K is the number of variables in the spectrum). The basic MSC consists of estimating two coefficients, a_i and b_i that ideally contain all the *physical* information in $\mathbf{z}_i$, based on the linear regression model

$$\mathbf{z}_i = a_i + b_i \mathbf{m} + \boldsymbol{\epsilon}_i \quad (1)$$

where $\boldsymbol{\epsilon}_i = [\epsilon_{i1}, \epsilon_{i2}, \dots, \epsilon_{ik}, \dots, \epsilon_{iK}]$ are the residuals that ideally contain all the *chemically* relevant information in $\mathbf{z}_i$, plus other unmodeled effects and random noise. Vector $\mathbf{m} = [m_1, m_2, \dots, m_k, \dots, m_K]$ is a common reference spectrum. After parameters a_i and b_i have been estimated, the corrected spectrum $\mathbf{z}_{i,\text{corrected}}$ for this sample is then generated by reversing Eq. 1, from the estimates of a_i and b_i , in an analogy to the univariate "reverse"² calibration:

$$\mathbf{z}_{i,\text{corrected}} = (\mathbf{z}_i - a_i)/b_i \quad (2)$$

These corrected spectra may be used as regressors in the subsequent multivariate calibration modeling of the analyte y_i from $\mathbf{x}_i = \mathbf{z}_{i,\text{corrected}}$ over a set of samples, $y_i = f(\mathbf{x}_i)$, $i = 1, 2, \dots$.

The common reference spectrum $\mathbf{m}$ in Eq. 1 may, for example, be defined as the **mean** of a set of N spectra of calibration samples:

$$\mathbf{m} = \frac{\sum_{i=1}^N \mathbf{z}_i}{N} \quad (3)$$

This reference spectrum $\mathbf{m}$ from the spectra of the N calibration samples $\mathbf{z}_i$, $i = 1, 2, \dots, N$ may also be applied to MSC of new spectra $\mathbf{z}_i$, $i = 1, 2, 3, \dots$ e.g., from future prediction samples; Eqs. 1 and 2 are the same for both kinds of samples.

In each calibration or prediction sample i , the unknown additive and multiplicative MSC coefficients a_i and b_i in Eq. 1 **may be estimated by ordinary least-squares regression of $\mathbf{z}_i$ on $\mathbf{m}$** , minimizing the sum of squared residuals in $\boldsymbol{\epsilon}_i$.

$$[a_i \ b_i] = ([1 \ \mathbf{m}']' [1 \ \mathbf{m}'])^{-1} [1 \ \mathbf{m}']' \mathbf{z}_i' \quad (4)$$

However, the process for estimating and correcting for scattering parameters a_i and b_i is only safe if the effects of chemical variation between $\mathbf{z}_i$ and $\mathbf{m}$ can be ignored; otherwise, the coefficients a_i and b_i may be contaminated with information about, e.g., the analyte, which will then be partially lost in $\mathbf{z}_{i,\text{corrected}}$ (Eq. 2). If applied to pure baseline separated absorbance bands, MSC (and basic EISC) will remove all relevant chemical information, as concentration will have a simple multiplicative effect on the spectral band. For this reason it is good practice to test the scatter coefficients a_i and b_i for information about the analyte or quality to be calibrated for. If a_i and b_i are found to be informative, they may even be included as additional regressor variables in the subsequent multivariate calibration models.

The problem of mixing chemical and physical information in the MSC may alternatively be reduced by down-weighting wavelength regions that carry chemical information. However, in some applications, like NIT of single wheat grains in the 850–1050 nm range, it is difficult to find wavelength regions that are sufficiently informative about the light scattering, but which do not carry chemical information. A more elegant approach would be an MSC model that includes information about the spectra of the chemical constituents.^{1,2,8} However, in some applications like the present one, the *in situ* constituent spectra are not known and are difficult to measure.

Basic Extended Inverted Signal Correction. The basic form of the EISC, ISC, is similar to MSC, but it may be more flexible and easier to understand for spectroscopists using multivariate calibration modeling by, for instance, PLSR. MSC, like its extensions, is based on a "reverse"⁸ correction in Eq. 2, compared to the model in Eq. 1. In contrast, the basic EISC, like its extensions, uses a "forward"⁸ model: the *same* direction of the relationship between spectrum $\mathbf{z}_i$ and reference spectrum $\mathbf{m}$ is kept, both in the model specification:

$$\mathbf{m} = a_i + b_i \mathbf{z}_i + \boldsymbol{\epsilon}_i \quad (5)$$

and in the final correction of the spectra:

$$\mathbf{z}_{i,\text{corrected}} = a_i + b_i \mathbf{z}_i \quad (6)$$

Hence, instead of regressing $\mathbf{z}_i$ on $\mathbf{m}$ in the model (Eq. 1) and then reversing this model in the signal correction step (Eq. 2), the inversed MSC (ISC/EISC) regresses $\mathbf{m}$

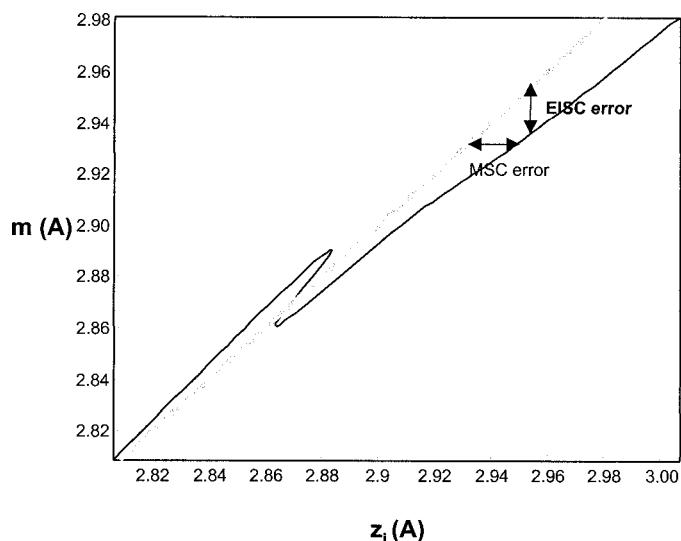


FIG. 1. Plot of a wheat kernel sample's input spectrum z_i vs. the mean (m) spectrum of the NIT wavelength range (850–1050 nm). While the MSC models the error horizontally on the individual spectra, the EISC models the error vertically on the average spectrum.

on z_i and uses this “forward” model directly in the signal correction step (Eq. 6).

The estimation of the parameters a_i and b_i may be done by ordinary least-squares regression

$$[a_i \ b_i] = ([1 \ z_i][1 \ z_i])^{-1}[1 \ z_i]'m \quad (7)$$

Like in MSC, weighted least-squares regression may be used instead if certain wavelengths are to be eliminated because of too-strong overlap between constituent spectra and light scattering effects. The rationale behind this model is explained by Martens et al.¹ The statistical difference between EISC and MSC in their basic form, discussed more theoretically by Helland et al.,⁵ is illustrated in Fig. 1. In the plot of reference spectrum m vs. a sample's input spectrum z_i , the residuals ε_{ik} are minimized horizontally in MSC (noise modeled on the individual spectra) and vertically in ISC/EISC (noise modeled on the average/reference spectrum).

General Spectroscopic Extensions of EISC. Just as the MSC can be extended into extended multiplicative signal correction (EMSC),⁸ the basic EISC can be extended to accommodate various types of physical or chemical *a priori* knowledge. In the present case, it is impossible to find wavelength ranges that distinguish the physical light scattering information from the chemical absorbance information. That must be expected to create problems for the MSC or basic EISC methods, but EMSC as well as EISC extended with chemical constituent spectra might solve the problems. However, contrary to the case in Martens et al.,¹ the present *in situ* chemical constituent spectra are not known. In the grain, water is probably bound to a greater or lesser extent to the protein, starch, and cellulose biopolymers, and the NIR *in situ* spectral contributions from the constituents may therefore be rather different from those of isolated constituents in a pure state.

On the other hand, the heterogeneity of the intact wheat grains may cause rather complex optical phenomena that are difficult to model explicitly, but which may

be approximated in more detail by polynomial extension of Eq. 5, e.g.,

$$m = a_i + b_i z_i + c_i z_i^2 + \epsilon_i \quad (8)$$

Moreover, we expect the light scattering coefficient to have some dependency on the wave number. The exponent of this dependency depends on particle size, which is unknown. A first order approximation of this is to include the wavelength vector λ , with polynomial terms, e.g.:

$$m = a_i + b_i z_i + c_i z_i^2 + d_i \lambda + e_i \lambda^2 + \epsilon_i \quad (9)$$

The parameters a_i , b_i , c_i , d_i , and e_i may be estimated by some sort of linear regression that makes the residual elements in ϵ_i small, with one separate model for each sample $i = 1, 2, \dots$. To ensure statistical and numerical stability, regression on standardized regressors was used, with a small ridge parameter.¹

The primary purpose of the EISC extension terms z_i^2 , λ , and λ^2 in Eq. 9 is to improve the estimation of the basic interference effects, the additive offset a_i (reflecting “baseline differences”) and the multiplicative slope b_i (reflecting “relative scatter coefficient differences”). But the extensions may also be used explicitly in the subsequent correction. Depending on whether or not one expects the corresponding coefficient estimates c_i , d_i , and e_i to carry information about the analyte, one may choose whether or not to use them in the subsequent correction. If they are thought (or found) to pick up irrelevant complexity from the data, the subsequent calibration modeling may be simplified after the EISC correction:

$$z_{i, \text{corrected}} = a_i + b_i z_i + c_i z_i^2 + d_i \lambda + e_i \lambda^2 \quad (10)$$

This is the EISC correction used in the present paper. Alternatively, if the extension coefficients d_i and e_i for the wavelength are expected to have picked up variation in the analyte that one does not want to lose, the effects $d_i \lambda$ and $e_i \lambda^2$ may be retained in the spectra by reducing the correction to

$$z_{i, \text{corrected}} = a_i + b_i z_i + c_i z_i^2 \quad (11)$$

Note that Eq. 8 is still a simple, linear (additive) model, but Eq. 9 works as a mixed additive/multiplicative pre-processing, in the sense that b_i is a multiplier that may reflect the relative scatter coefficient, while a_i may represent its additive baseline offset. In that sense the EISC correction (Eq. 10 or 11) is analogous to the correction by MSC (Eq. 2) and its extension.¹

MATERIAL AND METHODS

Samples. Wheat kernels (415) representing 43 different varieties or variety mixtures from two different locations in Denmark made up the calibration set, while wheat kernels (108) representing 11 different varieties from one location made up the test set.¹¹ All kernels were randomly chosen from bulk samples. The test samples were acquired with the calibration samples, but stored for about two additional months before measurement in order to provide a check for temporal drift in the samples and instrumentation. The NIT single seed data set is made available on the World Wide Web (Pedersen, Pram Nielsen, Munck & Engelsen, NITSingleSeed, www.models.kvl.dk).

Spectra Recordings. The single kernel transmittance spectra were collected on an Infratec 1255 Food and Feed Analyzer (Tecator AB, Höganäs, Sweden). Each kernel was placed in a single seed sample cassette, and transmittance spectra in the range 850–1050 nm were recorded. A tungsten lamp (50 W) and a diffraction grating were used to create monochromatic light. The light passed through the kernel, reaching the silicon detector in a diffuse pattern. Spectra were recorded three times for each kernel and the average of the three spectra was used for the calibrations. The time required for scanning (single scan) 23 single kernels in the cassette was about 90 s.

Protein Determination in Single Kernels. After the spectral recording of the intact wheat kernels each kernel was crushed in the single kernel characterization system (SKCS 4100, Perten Instruments Inc., Reno, NV) and the moisture content necessary for calculation of protein content in dry matter was determined. Subsequently, single kernel nitrogen content was determined directly by a modified Kjeldahl method.¹² Nitrogen in single kernel grits was transformed into ammonium sulfate by digestion (410 °C for 1 h) with 6 mL sulfuric acid (98%). The solution was then alkalized (25 mL 35% NaOH and 75 mL H₂O) and distilled into 25 mL boric acid (0.2%) with methyl red and bromocresol green indicator. The amount of resulting ammonia produced was determined by titration (0.0050 M HCl). The method is based on the assumptions that proteins contain 16 percent nitrogen and that non-protein nitrogen content can be neglected. The protein content is reported as 5.7 times the total nitrogen content for wheat kernels. This unusual calculation factor is due to the high nitrogen content of glutamine. Based on previous experience with samples of 30–40 mg of wheat flour, the analytical error of the analyte was expected to have an absolute standard uncertainty of 0.16% (percent protein content in dry matter).

Data Analysis. Multivariate data analysis was carried out using The Unscrambler version 7.6 (www.camo.com), except for the EISC calculations, which were programmed and carried out using MatLab version 6.1 (The Mathworks, Inc., Natick, MA). Conventional multivariate calibration models were developed from the 415 calibration samples using PLSR for protein content (y) from NIT spectra (X) after different types of spectral pre-transformation (MSC, basic EISC (Eq. 6), EISC with physically extensions (Eq. 10), second derivatives, and combinations of MSC and second derivatives). Optimal numbers of PLSR components (PCs), A_{opt} , as well as apparent root mean square error of Y-prediction, RMSECV, were estimated by cross validation within the calibration set. To ensure robust and representative segmentation in the cross validation, the 415 calibration samples were sorted for increasing value of protein content (y), and then split systematically into 10 cross-validation segments. Performance of calibration models was validated by predicting the protein content in the 108 samples (validation set), yielding the root mean square error of Y-prediction, RMSEP.

RESULTS AND DISCUSSION

Protein Content. The statistics of the Kjeldahl protein determination of the two sample sets are listed in Table

TABLE I. Means and standard deviations (SD) of single kernel protein data in the calibration and test sets.

Sample set	# of kernels	Mean protein (%)	SD (%)	Min. (%)	Max. (%)
Calibration	415	10.0	1.56	6.8	15.2
Test	108	9.8	1.75	7.0	17.0

I. The protein concentration in the calibration set ranges from 6.8 to 15.2%, while the concentration in the test set ranges from 7.0 to 17.0%. As indicated by the standard deviations in Table I, relatively few kernels have extreme protein content; however, a certain degree of extrapolation is required for the PLSR calibration model to cover the protein range of the test set. The higher protein content in the test samples is probably the result of a certain loss of moisture during the additional storage period.

Near-Infrared Transmittance Spectra. The NIT spectra of the single wheat kernels presented in this study cover the spectral region from 850 to 1050 nm in 2 nm steps containing primarily the second overtones of O–H (carbohydrates and water) and N–H (protein) stretching vibrations and the third overtone of the C–H (fats) stretching vibration. The fundamental O–H stretch for hydrogen bonded systems is typically found between 3400 and 3300 cm^{−1} (IR), corresponding to 2940–3030 nm, which will ideally give second overtones in the NIR region 980–1010 nm. Secondary amides (proteins) give rise to a fundamental N–H stretching vibration located near 3300 cm^{−1} (IR), corresponding to an ideal second NIR overtone near 1010 nm. Aliphatic C–H stretching vibrations are located between 3000 and 2840 cm^{−1}, corresponding to 3333–3521 nm, which will ideally give rise to third overtones in the Vis/NIR region between 833 and 880 nm. This spectral region is thus of utmost importance to food-related samples, as most of the important functional components are represented. The electromagnetic radiation is relatively high in energy, yet still absolutely nondestructive. Moreover, the absorption of the second and third overtones is much lower than the fundamental and first overtone vibrations, enabling larger sample volumes to be measured, which is very important when measuring heterogeneous systems.

Figure 2 displays raw NIT absorbance spectra of the 415 samples **in the calibration set**. From the figure, large additive offset and multiplicative scaling effects are readily observed. These are probably due to dispersive loss of light and changes in optical path length, caused by variations in kernel size and texture as well as kernel orientation in the sample cassette. These observed differences in light lost due to physical effects probably overshadow the absorbance changes due to concentration variations in the chemical constituent like starch, water, and protein.

Figure 3A displays the same NIT spectra after EISC pre-transformation **according to Eq. 10**. In comparison, Fig. 3B shows MSC pre-transformed spectra (Eq. 2), Fig. 3C shows second derivatives of the spectra followed by MSC, and Fig. 3D shows MSC-transformed spectra followed by second derivatives. Compared to the raw spectra in Fig. 2, all the calibration samples appear almost identical after the EISC. However, Fig. 4A shows that after **mean centering to remove the average spectral pat-**

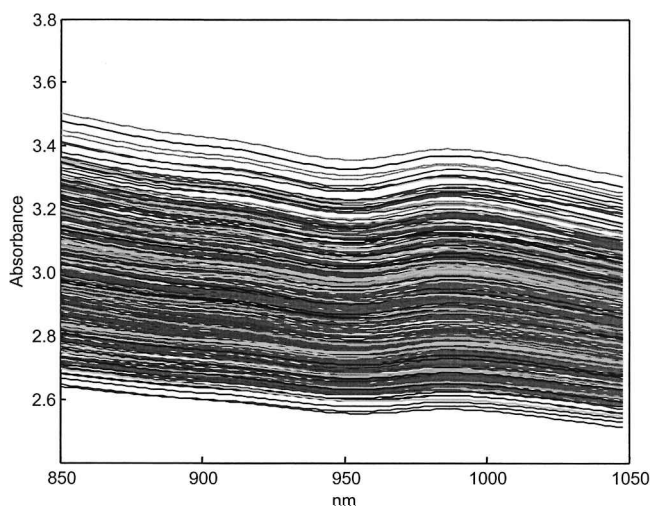


FIG. 2. Raw NIT spectra (850–1050 nm) of the 415 wheat kernels from the calibration set.

tern in the NIT data, the EISC pre-transformed spectra are quite different. Likewise, the mean-centered MSC pre-transformed spectra (Fig. 4B), mean-centered second derivatives of the spectra followed by MSC (Fig. 4C), and mean-centered MSC-transformed spectra followed by second derivatives (Fig. 4D) show clear differences between the samples. The question is whether these differences relate to variations in the protein content.

Figure 5 compares the calibration models from the raw (dotted) spectra (Fig. 2) and the EISC-transformed (solid) spectra (Fig. 3A), both after mean centering. It shows the regression coefficient summary for the two models obtained at a conservative model rank (i.e., 9 and 6 PCs, Fig. 5A) and at the number of PCs that appeared to be near optimal (11 and 7 PCs, Fig. 5B) judging from the cross validation. In general, the EISC has reduced the number of PCs required. Particularly in Fig. 5B, the two

predictors are relatively similar, although some differences can be observed. At the slightly lower rank in Fig. 5A the models are even more distinct.

Figure 6 compares the apparent performance of the raw NIT data and the EISC pre-transformation to various other pre-transformations for the single seed protein calibration models. The prediction errors (RMSECV) are plotted against the number of PLSR components. The figure reveals a significant reduction in the number of PLSR components needed, from the raw spectra to the pre-transformed spectra. Secondly and most interestingly, the plot reveals that only the EISC pre-transformation (solid) and the second derivative followed by MSC pre-transformation (densely dotted) are able to provide an optimal model according to the level of the prediction error in the calibration set.

The prediction error in the calibration set for the PLSR model based on EISC-transformed spectra was estimated by cross validation to 0.49% protein (7 PCs). The corresponding prediction error for the PLSR model based on the second derivatives followed by MSC-transformed spectra was also estimated to 0.47% protein (5 PCs), while PLSR models based on raw, differentiated, basic EISC or MSC-treated spectra never reached a prediction error less than 0.55% protein, regardless of the number of PLSR components applied. The improved prediction performance agrees with the findings of Delwiche,¹⁰ who showed that the two-step procedure of using second derivatives followed by MSC gave a better single seed NIT model for prediction of protein content. The most significant result of this comparison is that the single-step EISC performs equally as well as the double transformation, but it is perhaps also noteworthy that basic EISC (ISC) performs just as the sister algorithm MSC on the calibration set, but with a significantly better result on the test set.

The two-step method based on second derivatives fol-

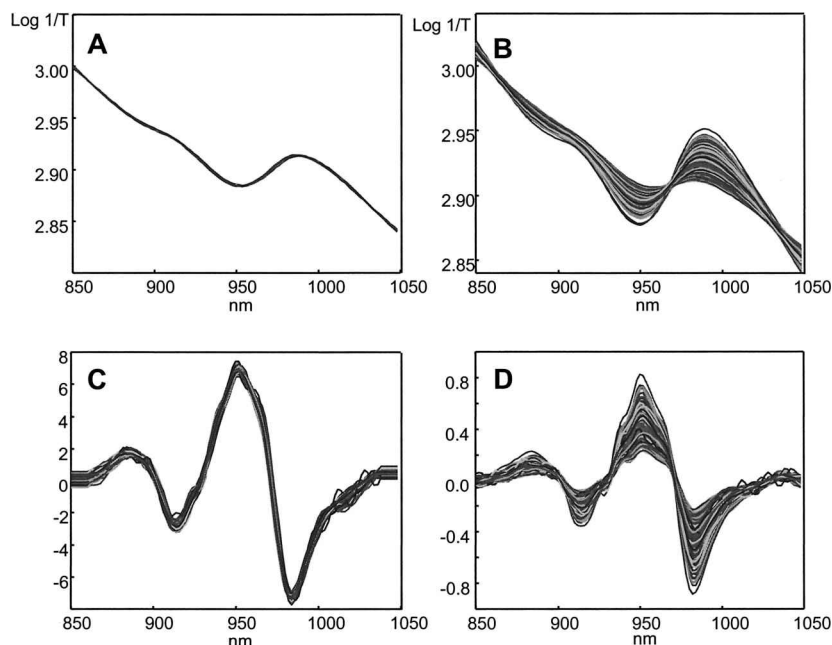


FIG. 3. NIT spectra (850–1050 nm) of the 415 wheat kernels from the calibration set; (A) EISC transformed (Eq. 10), (B) MSC transformed (Eq. 2), (C) second derivatives followed by MSC, and (D) MSC followed by second derivatives.

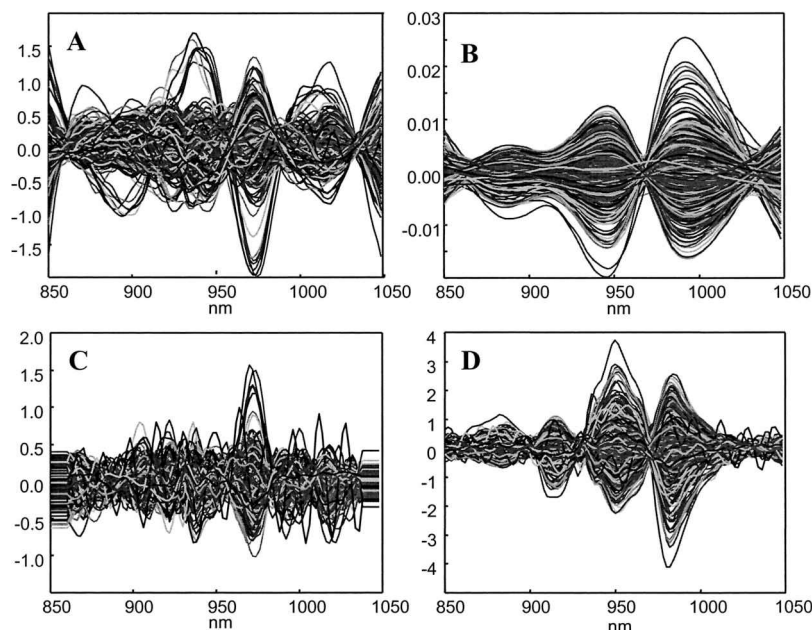


FIG. 4. Mean-centered NIT spectra (850–1050 nm) of the kernels from the calibration set; (A) EISC transformed, (B) MSC transformed, (C) second derivatives followed by MSC, and (D) MSC followed by second derivatives.

lowed by MSC-corrected spectra performs considerably better than the opposite two-step method, MSC followed by second derivatives (Fig. 6). This emphasises that the order of the applied pre-transformations is important and that conventional MSC is a poor model when the scatter is not linear.¹³ In the MSC it is assumed that the scatter is linear throughout the spectral range, since the whole spectrum is linearly adjusted by one slope and one offset. However, if the loss of light due to light scattering and other effects is not this simple, the MSC correction is not suitable and, consequently, a model based on MSC followed by the second derivatives will not be optimal. On the contrary, by using the second derivatives, it appears

that the spectra are successfully corrected for local offsets and linear trend variations.

Validation. The test set (108 single wheat kernels representing 11 of the varieties included in the calibration, but stored for an additional two months) was measured on the same NIT instrument, analyzed for protein content by the same method, and used for testing the (long-term) stability of various calibration models. Figure 7 shows the EISC parameter estimates according to Eq. 9, the coefficients a_i (additive offset), b_i (multiplicative scaling), c_i (for squared spectrum), d_i (for wavelength), and e_i (for squared wavelength) for the 415 calibration samples as well as for the 108 test samples. Figure 7F illustrates that the samples have been sorted for increasing protein content (y) within each of the two sample sets for simpler

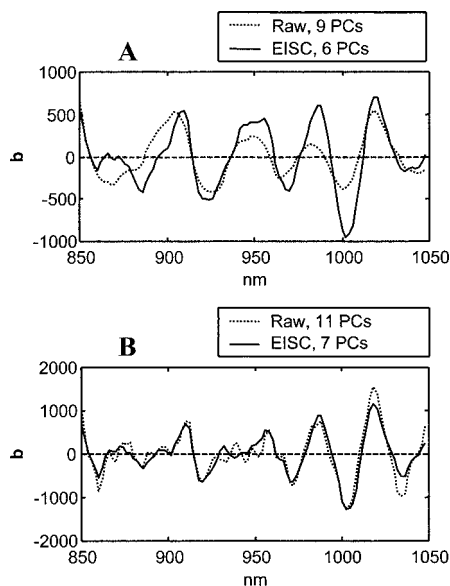


FIG. 5. The regression coefficients for the calibration of raw input spectra (.....) for 9 PCs (A) and the optimal 11 PCs (B), and for the calibration of EISC-transformed spectra (—) for 6 PCs (A) and the optimal 7 PCs (B).

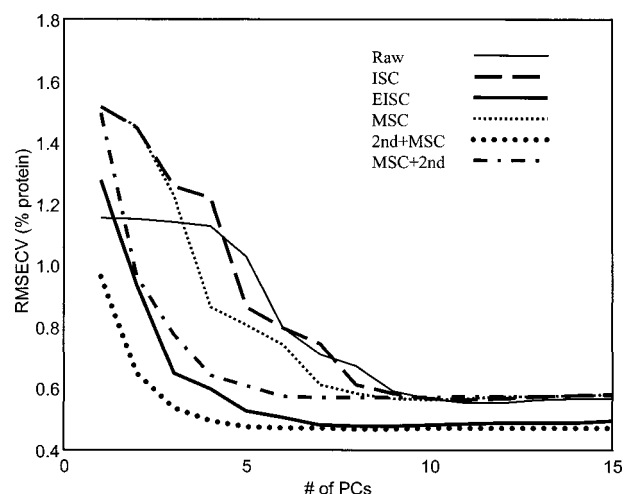


FIG. 6. RMSECV vs. PLSR components (PCs) for models based on different pre-transformed NIT spectra; raw spectra, ISC-corrected spectra, EISC-corrected spectra, MSC-corrected spectra, spectra transformed to the second derivative followed by MSC correction (2nd + MSC) and MSC correction followed by second derivative (MSC + 2nd).

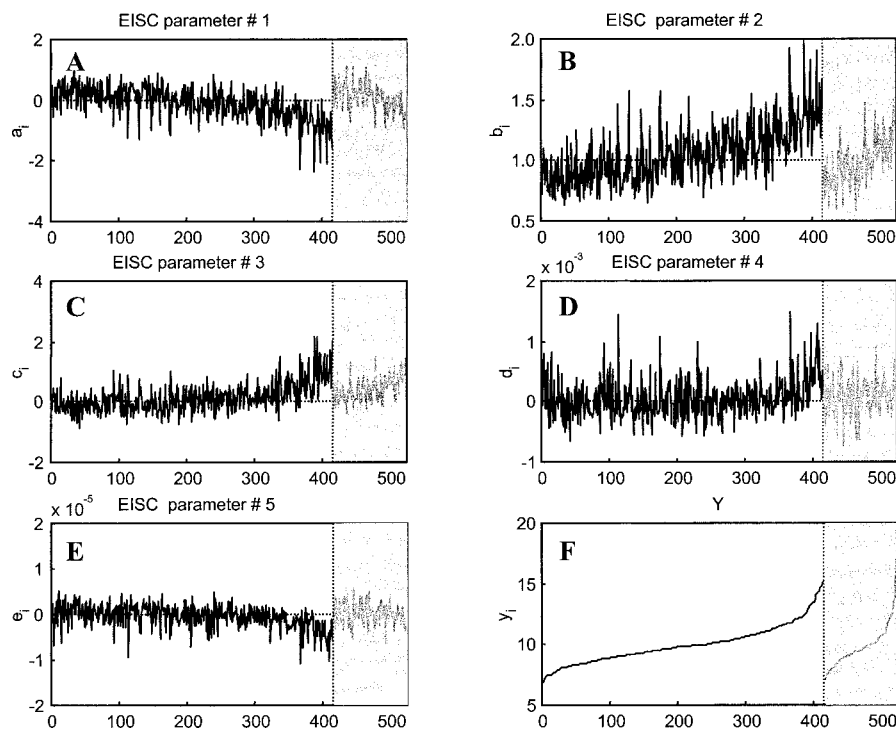


FIG. 7. The estimated EISC parameters (Eq. 9) plotted against sample i for the 415 calibration samples (black) as well as for the 108 test samples (grey), sorted according to increasing protein content in the two data sets; (A) EISC parameter # 1: a_i (additive/offset); (B) EISC parameter # 2: b_i (multiplicative/relative scatter scaling of input spectrum); (C) EISC parameter # 3: c_i (effect of squared spectrum); (D) EISC parameter # 4: d_i (effect of wavelength); and (E) e_i (effect of squared wavelength). The increasing protein content y_i within each of the two sample sets is shown in subplot F.

cross validation in the calibration set and simpler visual interpretation of the EISC parameters in both sets. The EISC parameters (Figs. 7A–7E) show highly erratic variations, especially in b_i (multiplicative scaling) and d_i (wavelength). But some systematic changes with the protein content may be observed in both sets, particularly at the highest protein levels (>11%). This is an indication that the EISC may have picked up and removed some variation related to the analyte. The cause and nature of this lost analyte information is unclear, but needs to be studied in more detail.

Figure 8 compares the calibration set and the test set, before and after the EISC. Spectroscopically, the mean-centered spectra of the test samples appear normal in the raw data (Fig. 8B), compared to the calibration set (Fig. 8A), while they show a very distinct pattern after the EISC (Fig. 8E vs. Fig. 8D). This is an indication that all test samples deviate in the same systematic way from the calibration mean spectrum $\mathbf{m}$. If this type of deviation is also present among the calibration samples, it may be modeled and corrected for in the calibration model; if not, the systematic deviations will cause grave errors in the predicted percent protein in the test set. The peaks just below 950 nm in Fig. 8D indicate that some of the calibration samples indeed display the same general pattern, but this needs to be verified in the prediction of protein.

Figures 8C and 8F compare the predictive performance before and after EISC. The long curves show the estimated error for protein content y predicted from the 100 NIT wavelength channels $\mathbf{X}$ for PLSR calibration models using between 0 and 15 PCs, for the cross-validated calibration set (RMSECV, solid) and the test set (RMSEP,

dashed). When using the raw spectra in Fig. 8C as $\mathbf{X}$, the cross validation shows that several of the first PCs (2, 3, 4, 5) have little or no predictive relevance for the protein content; hence, they must reflect very strong covariance structures in the NIT spectra. More importantly, the predictive ability in the test set changes erratically with the increasing number of PCs; obviously, a wrong choice in the number of PCs to be used for prediction may cause very high prediction errors in the test set.

In contrast, when using the spectra after EISC in Fig. 8F as $\mathbf{X}$, the cross-validation curve falls smoothly, as desired. The model is shown to require at least 4 PCs. The test set curve is very similar to the cross-validation curve after 4 PCs.

The two short curves in Fig. 8F show the estimated prediction errors using instead the 5 EISC parameters [a_i, b_i, c_i, d_i, e_i] (Eq. 9) from the different samples as $\mathbf{X}$, instead of the 100 wavelength channels. Some predictive ability for the protein content (y) is evident in the cross-validation curve (squares). Hence, the EISC may have removed Y-relevant information. However, in the test set (diamonds) the predictive ability for y is not as good. Attempts (not shown here) at joining the NIT data with the EISC parameters as extra variables, $\mathbf{X} = [\mathbf{z}_{i,\text{corrected}}, a_i, b_i, c_i, d_i, e_i]$, using weighted least-squares PLSR, gave a slight but insignificant improvement in RMSECV (calibration set), but no improvement in RMSEP (test set). It thus appears that the “physical” information apparently removed by the EISC in these data was not important or reliable for the prediction of chemical protein content: the errors that they contribute to the calibration model

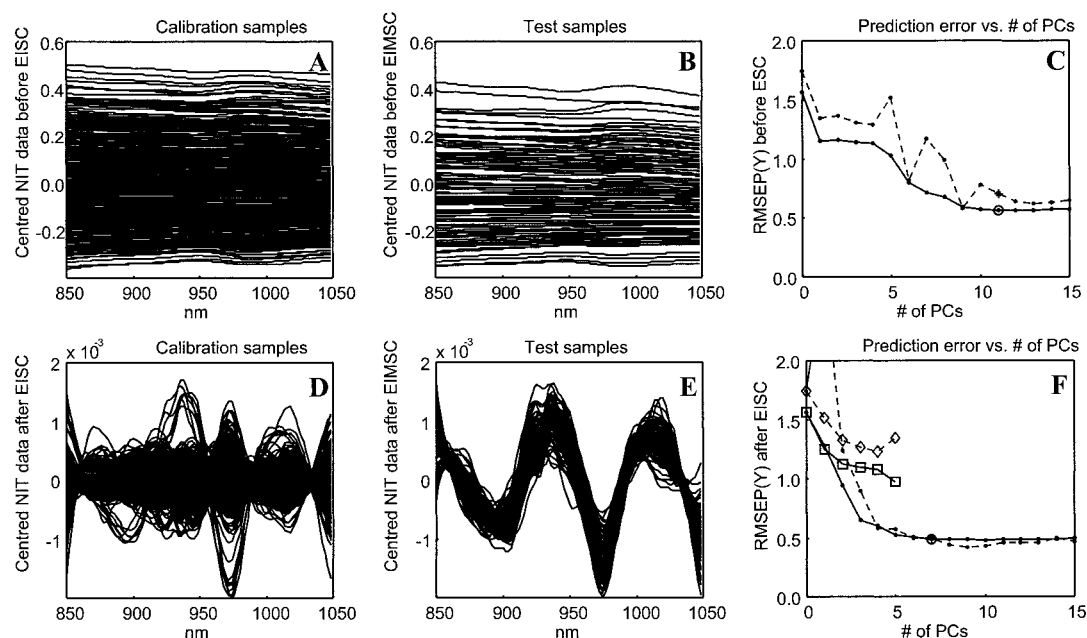


FIG. 8. The mean-centered NIT spectra for (A) the calibration samples and (B) the test samples, and the EISC transformed mean-centered NIT spectra for (D) the calibration samples and (E) the test samples. Prediction error vs. the number of PCs for the calibration samples (—) and the test samples (---) (C) before EISC transformation and (F) after EISC transformation. The two short curves in subplot F show the prediction errors calibrating only with the five EISC parameters (Eq. 9), $\mathbf{X} = [a, b, c, d, e]$; the calibration samples (—) and the test samples (---).

are greater than the otherwise unmodeled Y-variation that they can remove.

The calibration and the test set results for all the tested pre-transformation methods are summarized in Table II in terms of the RMSECV (calibration set) and RMSEP (test set) read at the optimal number of PCs, and of the correlation coefficients based thereon. Compared to the untransformed raw data, the basic EISC/ISC did not affect the results very much. However, there is an improved correlation, both for the calibration set (from 0.93 to 0.95) and for the test set (0.96 to 0.98), after applying the full EISC (Eq. 10) to the NIT spectra. The protein calibration model predicts the test kernels well throughout the protein range; the prediction error (RMSECV and RMSEP) ends as low as 0.49% protein in a protein range of 7 to 17%. This RMSE level approaches the sampling and measurement error on the single seed protein determination (0.16% determined for samples of 30–40 mg of flour), and the results demonstrate a very good and robust protein calibration on single wheat kernels.

TABLE II. Performance statistics of the PLSR models for single seed protein predictions using single seed NIT spectra from the calibration set (415 kernels) and the subsequent test set (108 kernels). CV is cross validation, RMSECV is the root mean square error of cross validation, and RMSEP is the root mean square error of prediction.

Pre-transformation	# of PLSR components	Correlation		Prediction error (% protein)	
		Cal. set (CV)	Test set	Cal. set RMSECV	Test set RMSEP
Raw	11	0.93	0.96	0.55	0.70
ISC	9	0.93	0.95	0.58	0.69
EISC	7	0.95	0.98	0.49	0.49
MSC	9	0.93	0.95	0.57	0.78
2nd + MSC	5	0.95	0.98	0.47	0.48
MSC + 2nd	7	0.92	0.93	0.60	0.66

The conclusion is that the basic ISC performs equally as well as the traditional MSC, perhaps even slightly less aggressively on the calibration set, resulting in an improved test set prediction. Both the EISC with general (physical) extensions and the two-step “second derivatives followed by MSC” in this data set can correct for spectra interferences that are not corrected by the more “classical” pre-transformations, MSC or second derivatives. The EISC is particularly promising because it is more flexible and easier to understand than the “classical” MSC and two-step methods. In this study we have emphasized a general applicable version of the EISC, but its flexible approach allows simple implementation of system-specific interferences such as known analytes.¹ In a future implementation, we will work on a version in which the correction coefficients are constrained to be orthogonal to the reference value $\mathbf{y}$, with the aim of optimizing subsequent regression models with even less loss of analyte information.

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